Alexandre Kallen

Materials Engineer, Master's Graduate in Plasma Physics



Profile

Recent graduate in plasma physics and materials engineering, with solid skills in modelling, data analysis, thermodynamics and materials science. Eager to bring this rigorous, versatile scientific approach to technical, industrial and R&D projects, and open to pursuing a PhD.

Contact

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Category B licence, own vehicle

Thionville, Lorraine, France

LinkedIn

Languages

French :	Native
English :	**C1** Level
German :	**B2** Level
Spanish :	**A2** Level
Chinese :	**HSK3** Level

Skills

- Numerical modelling
- Data analysis
- Microsoft Office, report writing
- Materials science
- Teamwork

References

Dr. Roch SMETS
Associate Professor | LPP – CNRS
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Dr. Fabien BERNACHY BARBE
Research Engineer | CEA
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PROFESSIONAL EXPERIENCE

Laboratoire de Physique des Plasmas (LPP)
Palaiseau, France

R&D Intern, Modelling and data analysis
Mar. 2026 – Aug. 2026

Numerical study of forced magnetic reconnection in the solar wind.

- Ran numerical simulations and parametric studies to analyse the behaviour of complex physical systems.
- Processed, analysed and visualised large datasets using Python on a computing cluster, in order to identify trends and compare results across parameter sets.

Commissariat à l'Énergie Atomique (CEA)
Cadarache, France

R&D Intern, Modelling and data analysis
Feb. 2024 – Aug. 2024

Modelling caesium migration in fuels for sodium-cooled nuclear reactors.

- Developed a Python model to simulate fission product transport in materials for next-generation nuclear reactors.
- Analysed data and studied the influence of various parameters to validate and improve the models developed.

Universitat Politècnica de València (UPV)
València, Spain

R&D Intern, Materials optimisation
Sep. 2023 – Feb. 2024

Development of graphene-based porous materials for water treatment.

- Optimised manufacturing parameters to study their influence on material properties and performance.
- Analysed and interpreted experimental results obtained from various characterisation techniques, in order to link manufacturing parameters to material properties.

CERATIZIT Luxembourg
Mamer, Luxembourg

Process Intern, Process optimisation
May 2021 – Jun. 2021

Optimisation of a method for extracting the organic components of tungsten carbide powders before their industrial processing.

- Studied and optimised the process parameters (temperature, treatment time and solvents used) in order to improve the performance of the extraction.
- Carried out the analysis and separation of the organic components present in the powders, prior to their industrial processing.

EDUCATION

Master's in Physics
Nancy, France

Specialisation: plasma physics and fusion energy
Sep. 2025 – Sep. 2026

Université de Lorraine, Faculté des Sciences et Technologies

École Européenne d'Ingénieurs en Génie des Matériaux
Nancy, France

Materials engineering, glass and ceramics specialisation
Sep. 2021 – Sep. 2024

Université de Lorraine, EEIGM

Cycle Préparatoire Polytechnique
Nancy, France

Preparatory classes for engineering schools, physics-chemistry and mechanics
Sep. 2019 – Sep. 2021

Université de Lorraine, CPP